

Research papers

A process-driven deep learning hydrological model for daily rainfall-runoff simulation

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ABSTRACT

Although deep learning (DL) models, especially long-short-term memory (LSTM), demonstrate greater accuracy than process-based models in rainfall-runoff simulation, the predictions from process-based models are more physical than DL models. The main reason is that DL models have almost no process understanding capabilities like process-based models beyond their fitting capability. In this study, we developed a process-driven DL model under a unified DL architecture to improve the process awareness of DL models. To implement the model, a conceptual hydrological model (EXP-HYDRO) is implanted into a recurrent neural network (RNN) cell as a process driver for providing multi-sub-process variables related to the runoff process, and an Entity-Aware LSTM (EA-LSTM) cell is incorporated as a post-processor layer, resulting in the Process-driven RNN-EA-LSTM (PRNN-EA-LSTM). Under the assistance of the process driver, the model performance of PRNN-EA-LSTM on the 531 catchments from the Catchment Attributes and Meteorology for Large-sample Studies dataset is more robust than the pure DL model, and better than using only EXP-HYDRO as the input of EA-LSTM (i.e., EXP-HYDRO-EA-LSTM). Specifically, the median Nash-Sutcliffe efficiency (NSE) of PRNN-EA-LSTM in local and regional simulation is 0.03 and 0.02 higher than LSTM and 0.01 higher than EXP-HYDRO-EA-LSTM. Additionally, PRNN-EA-LSTM significantly enhances the low flow simulations and reduces the catchments number with negative NSE. This study demonstrates that process-based models can help DL models better represent the rainfall-runoff relationship under a unified architecture. Consequently, integrating the adaptability of process-based models into the DL architecture is anticipated to bolster the process understanding capabilities of DL models.

1. Introduction

1.1. Process-based models in hydrology

Hydrology is a cornerstone discipline in the geosciences, and rainfall-runoff simulation has always been a key focus of research within hydrology (Beven, 2012). Since the nineteenth century, differential Equations (DE) are extensively employed by hydroscintists for studying various hydrological phenomena and processes such as evapotranspiration and snowmelt, baseflow and surfaceflow (Fatichi et al., 2016; Bauer et al., 2015). With the ongoing advancement in understanding and updating of hydrological processes, researchers have gradually formed these DE into a series of relatively complete and well-

interpretable process-based models. These models have been used to predict and explain various hydrological tasks, especially the prediction of floods (Hostache et al., 2018) and drought disasters (Kang and Sridhar, 2017).

Generally, process-based models describe the runoff process within a catchment while also describing other sub-processes (e.g., soil water, snowmelt, evapotranspiration, baseflow, and groundwater, etc.) (Kumar, Samaniego, & Attinger, 2013). These sub-processes help to provide a full narrative of the event (Berghuijs et al., 2016). For example, high antecedent soil moisture or thawing snow primed the catchment for floods, and runoff from catchments in a long-term drought state is mainly supplied by baseflow generated by water storage in the catchment. Therefore, including these sub-processes greatly improves

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the runoff process understanding of models (Feng et al., 2022).

However, process-based models still suffer from a dichotomy between interpretability and simulation accuracy (Nearing et al., 2021). The problem currently is that process-based models degrade significantly in performance when calibrated for multiple catchments together instead of for a single catchment alone. Some researchers have sought to develop parsimonious models that should be as simple as possible for maintaining interpretability and complex enough to achieve high simulation accuracy (Höge et al., 2018; Gharari et al., 2021). Often, only a few model states and processes are sufficient as a valid theory to describe a specific event (Fenicia et al., 2016). While such simplified models achieve certain results, their simulation accuracy usually only reaches the threshold of good performance (Nash Sutcliffe efficiency (NSE) ≥ 0.55 is considered the threshold of good performance, Knoben et al., 2019; Newman et al., 2015).

1.2. Deep learning in hydrology

In the context of Artificial Intelligence's global success and emergence of big Earth data, deep learning (DL) models, in particular long-short-term memory (LSTM) neural networks (Hochreiter and Schmidhuber, 1997), have frequently outperformed process-based models in many hydrological tasks (Jahangir, You & Quilty, 2023; Wunsch et al., 2021), especially rainfall-runoff simulation (Kratzert et al., 2019a,b; Gholizadeh et al., 2023). In Hydrology, the first demonstration of DL outperforming a process-based model that we are aware of was by Hsu et al. (1995), who compared a calibrated Sacramento Soil Moisture Accounting Model (SAC-SMA) with a feed-forward artificial neural network across a range of flow regimes. For LSTM, Kratzert et al. (2019a,b) showed that regionally trained LSTM network outperforms catchment-specific calibrations of several process-based models. Additionally, the researchers focused on improvements to the standard LSTM architecture and proposed an Entity-Aware LSTM (EA-LSTM, Kratzert et al. (2019a,b)) for rainfall-runoff modeling, which can clearly distinguish different or similar rainfall-runoff behaviors among different catchments and outperforms five process-based models. Currently, EA-LSTM stands out as one of the top-performing DL models, especially for regional simulations, with a median NSE of 0.74 using the ensemble. Consequently, it is widely publicized that LSTM represented a "step-change" in performance, indicating that process-based models were far from optimality (Nearing et al., 2021).

While it is recognized that DL models outperform process-based models in terms of predictive performance in rainfall-runoff simulation, geoscientists remain cautious when using DL models. DL models tend to focus on learning what they observed hydrological processes, thereby reacting to mismatches between predictions and observations, which making it difficult to proactively understand the underlying physical dynamics involved in the learned processes (Blöschl et al., 2019; Karpatne et al., 2019). This easily leads DL models to make catastrophic predictions in unknown areas, or even learn false hydrological relationships that appear accurate based on the training data but fail to generalize well to new datasets (Sit et al., 2020; Shen & Lawson, 2021). Such lessons are not uncommon in Earth System Science (Reichstein et al., 2019). The main reason is that it is difficult for DL models to address hydrodynamic problems from within the model, similar to process-based models. As a result, the absence of physical process elements prevents DL models from perceiving potential physical dynamics, thereby limiting their ability to achieve more profound hydrological applications.

1.3. Synergy of physics and deep learning

From the above developments, a question arises as to how to improve the process understanding of DL models in hydrological applications and thereby increase the robustness of predictions (Nearing et al., 2021). Considering the respective merits of process-based and DL models,

hybrid modeling strategies have become the most effective method to respond to this scientific problem (Konapala et al., 2020; Willard et al., 2022). Some feasible hybrid modeling strategies have been actively explored, such as introducing physical constraints in the loss function of DL models to learning and correcting mismatched relationships between physical constraints and observations (Xie et al., 2021; Karpatne et al., 2017), or adding physical variables of interest output by process-based model to the training samples to improve the robustness and physical consistency of the DL model predictions (Jia et al., 2021; Yang et al., 2020). These strategies have been successfully applied to rainfall-runoff simulation and river network flow and temperature predictions. In comparison, a further vision is to enable DL models to incorporate process elements, enabling them to proactively capture the underlying physical dynamics associated with the learned hydrological processes, thereby improving the model's process understanding. (Reichstein et al., 2019).

Recently, a novel DL modeling framework with enhanced process understanding has emerged, which embeds a daily time-step spatially lumped hydrological model (EXP-HYDRO, including sub-processes such as snowmelt, evapotranspiration, baseflow and surfaceflow, Patil & Stieglitz, 2014) into a common recurrent neural network (hereafter called EXP-HYDRO RNN, Jiang, et al., 2020). EXP-HYDRO RNN outputs runoff in a process-based way. This promising DL modeling framework has garnered significant attention from researchers, leading to the proposal of several variations and adaptations (Hoge et al., 2022; Li et al., 2023). However, since the computational logic of DL architecture is completely replaced by a process-based model, its predictive performance on a large sample dataset (Catchment Attributes and Meteorology for Large sample Studies (CAMELS, Addor et al., 2017)) is greatly affected (refer to EXP-HYDRO RNN, the median NSE of 0.48 and 0.51 in local and regional modeling). Nonetheless, Jiang et al. (2020)'s research remains a significant stride towards the realization of DL models with enhanced process understanding.

However, to the best of our knowledge, few researchers have recognized the additional value of EXP-HYDRO RNN. In fact, it not only incorporates process elements inside the DL model, but also enables the physical variables of interest to be output in a process-based way. This feature enables EXP-HYDRO RNN to serve as a process driver within a unified DL architecture. By incorporating the physical variables of interest as feedback into the network, it holds the potential to enhance the comprehension of multi-hydrological processes, ultimately improving the understanding of runoff processes. For example, in arid catchments, runoff is primarily sustained by baseflow, which is generated through water storage to maintain a normal hydrological cycle during extended dry periods. This is difficult to achieve by pure DL models that lack baseflow dynamics. Considering the latest advances in hybrid modeling strategies, this study combined an EXP-HYDRO RNN cell with an EA-LSTM cell to implement a Process-driven RNN-EA-LSTM (called PRNN-EA-LSTM) model. Within PRNN-EA-LSTM architecture, the former endows the physical mechanisms and underlying dynamics, while the latter fills gaps in physical knowledge through a data-driven approach (e.g., some rainfall-runoff processes that cannot be solved analytically by EXP-HYDRO RNN alone). PRNN-EA-LSTM strives to achieve state-of-the-art hydrological simulation accuracy while improving predictions robustness and physical consistency. Specifically, we pay attention to three aspects about PRNN-EA-LSTM: (1) Can PRNN-EA-LSTM achieve simulation accuracy close to or even surpass EA-LSTM, especially in the regional simulation?; (2) Can the physical variables of interest educate the model to learn underlying physical dynamics when they are passed into the network?; (3) Synergy of physics and DL in PRNN-EA-LSTM.

The remainder of this paper is organized as follows: Details on the CAMELS dataset are described in Section 2. Section 3 describes the methodology for PRNN-EA-LSTM, followed by results in Section 4. Lastly, conclusions and outlook are drawn in Section 5.

2. Data

The data used in this study were obtained from the open-source Catchment Attributes and Meteorology for Large-Sample Studies (CAMELS) dataset in the conterminous United States (CONUS), (Addor et al., 2017). The CAMELS dataset includes the area-averaged daily meteorological forcings and catchment attribute data (including climate, topography, vegetation, soil, and geological characteristics) for 671 catchments across the CONUS with limited human disturbance (Table 1). These catchments are largely natural and have long gauge records (1980–2010) available from the United States Geological Survey National Water Information System. In this study, we used only 531 of 671 CAMELS catchments—these were the same catchments that were used for model benchmarking by Newman et al. (2017), who removed catchments from the full CAMELS dataset with (i) large discrepancies between different methods of calculating catchment area and (ii) areas larger than 2,000 km².

3. Methodology

3.1. A brief overview of PRNN-EA-LSTM

The proposed PRNN-EA-LSTM architecture is shown in Fig. 1, which mainly consists of two input blocks (i.e. meteorological forcings and catchment attribute data), two neural network cells, and an output block (i.e. system responses). First, a common RNN cell is used to embed a

Table 1

Table of meteorological forcing and catchment attribute data used in this study.

Meteorological forcings	
T	2 m daily mean air temperature (°C)
L _{day}	Day length (hour)
RI	Average daily precipitation (mm/day)
SRAD	Surface-incident solar radiation (W/m ²)
Vp	Near-surface daily average (Pa)
Catchment attribute data	
P _{mean}	Mean daily precipitation
PET _{mean}	Mean daily potential evapotranspiration
Aridity index	Ratio of Mean PET to Mean Precipitation
P _{seasonality}	Estimated by representing annual precipitation and temperature as sin waves Positive (negative) values indicate precipitation peaks during the summer (winter). Values of ~0 indicate uniform precipitation throughout the year
Frac _{snow_daily}	Fraction of precipitation falling on days with temp < 0 °C
High _{prec_freq}	Frequency of days with $\leq 5 \times$ mean daily precipitation
High _{prec_dur}	Average duration of high precipitation events (number of consecutive days with $\leq 5 \times$ mean daily precipitation)
Low _{prec_freq}	Frequency of dry days (<1 mm/day)
Low _{prec_dur}	Average duration of dry periods (number of consecutive days with precipitation < 1 mm/day)
Elev _{mean}	Catchment mean elevation
Slope _{mean}	Catchment mean slope
Area	Catchment area
Frac _{forest}	Fraction of catchment covered by forest
LAI _{max}	Maximum monthly mean of leaf area index
LAI _{diff}	Difference between the max. and min. mean of the leaf area index
GVF _{max}	Maximum monthly mean of green vegetation fraction
GVF _{diff}	Difference between the maximum and minimum monthly mean of the green vegetation fraction
Soil _{depth} (Pelletier)	Depth to bedrock (maximum 50 m)
Soil _{depth} (STATSGO)	Soil depth (maximum 1.5 m)
Soil _{porosity}	Volumetric porosity
Soil _{conductivity}	Saturated hydraulic conductivity
Max _{water_content}	Maximum water content of the soil
Frac _{sand}	Fraction of sand in the soil
Frac _{silt}	Fraction of silt in the soil
Frac _{clay}	Fraction of clay in the soil
Frac _{carb_rocks}	Fraction of the catchment area characterized as “carbonate sedimentary rocks.”
Geol _{permeability}	Surface permeability (log10)

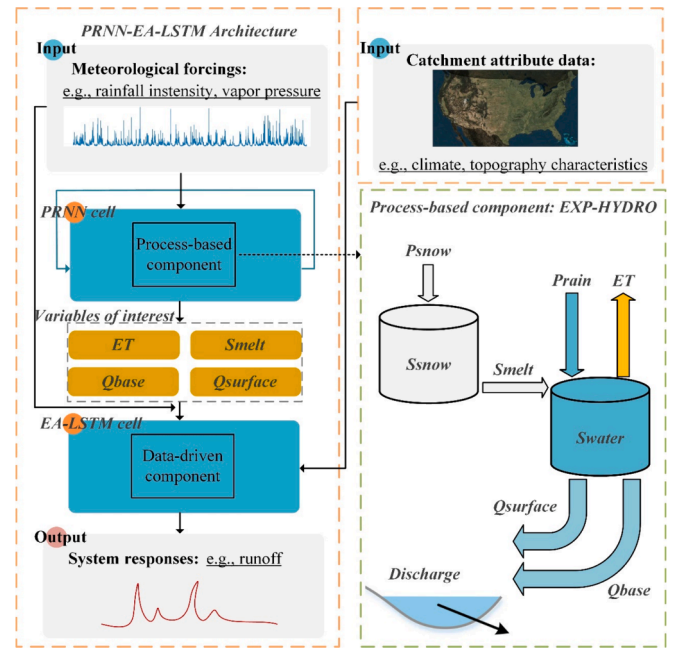


Fig. 1. The PRNN-EA-LSTM architecture for rainfall-runoff simulation.

process-based component between the input blocks and the output block so that the model includes the water balance mechanism and physical sub-processes described by EXP-HYDRO. EXP-HYDRO uses the state variables S_{snow} and S_{water} to maintain the water balance, and uses 6 physical sub-processes to describe the runoff process (See Appendix A1 for EXP-HYDRO details). Next, the PRNN cell outputs the variables of interest in a process-based way, and passes these variables together with the input blocks data to an EA-LSTM cell to participate in the training of the entire network. Finally, the trained PRNN-EA-LSTM outputs runoff.

For PRNN cell, which is analogous to a common RNN cell, the backbone of the PRNN cell is formed by recurrent cells that retain memories of the preceding sequence. Within the recurrent cells, the connections between neurons (input x , state s and output y) are specified by the ordinary differential equations representation in an explicit discrete form and the neurons parameters (i.e., weights and biases) are replaced by physically meaningful parameters $p \{f, S_{max}, Q_{max}, D_f, T_{max}, T_{min}\}$ (see Table A1 for the meaning and value range of each parameter) in EXP-HYDRO. PRNN cell can therefore diverse various variables of interest and contribute valuable physical knowledge to the entire network. For the specific implementation details of this cell, this study refers to Jiang et al., (2020). Table A2 summarizes the implementation logic (pseudocode) of the PRNN cell.

For EA-LSTM cell, the core idea is to adjust the input gate i to a function that remains constant over time, while keeping the other functions (i.e., f, o, g) unchanged. This adjustment enables the network to clearly differentiate between distinct or similar hydrological behaviors (here, refers to the rainfall-runoff process) across various catchments, making it particularly advantageous for cross-catchment learning. In addition to enhancing the capability to fit sequences, the data-driven component also allows static data related to catchment attributes (e.g., climate and terrain properties) to be input to the model to improve the transferability in cross-catchment learning (i.e., regional simulation). Table A3 summarizes the implementation logic (pseudocode) of the EA-LSTM cell. Notably, the EA-LSTM does not include static input in local modeling and can thus be regarded as the standard LSTM, so the EA-LSTM and its variations in the following local modeling cases are referred to as the LSTM or LSTM variants.

PRNN-EA-LSTM contains two types of parameters: the physically meaningful parameters $p \{f, S_{max}, Q_{max}, D_f, T_{max}, T_{min}\}$ of EXP-HYDRO

and the network weight parameters w {weights and biases} of EA-LSTM. Including w greatly increases the number of parameters that need to be estimated, which increases the complexity of training. This study achieves simultaneous estimation of p and w during the training process by minimizing the loss function between the data and the output of the model, and finally the PRNN-EA-LSTM is jointly parameterized by the two.

3.2. Experimental design

3.2.1. Model Training, Hyperparameters, loss function and other details

Daily precipitation P , daily temperature T and day length L_{day} are EXP-HYDRO inputs and the daily runoff at the catchment outlet is the output of interest. This model involves six physically meaningful parameters that control hydrological behavior. We implemented embedding EXP-HYDRO into a common RNN cell on TensorFlow, and other platforms can work similarly.

For local PRNN-LSTM, first, the PRNN cell takes P , T , L_{day} , $SRAD$ and V_p (shape [batch size, sequence length, 5 (input feature dimensions)]) as inputs, providing the physical variables of interest ET , $Smelt$, Q_{surface} and Q_{base} (together with 5 inputs sequence, new shape [batch size, sequence length, 9 (input feature dimensions)]) is used for LSTM cell (two-layer hidden layer, hidden size is 40). For sequence data, regional PRNN-EA-LSTM is the same as the former, inputting the input vector with the shape [batch size, sequence, 9 (input feature dimensions)] to the f , o , and g gate functions of the EA-LSTM cell (one hidden layer, hidden size is 256), and separately separate the 27 catchment attribute data input to the i gate functions that remain constant over time. Both types of models mentioned above use the Adam algorithm for training the model, with a learning rate of 0.01 for parameter estimation of w and p . During training, the maximum iteration step size is set to 200. Considering computational efficiency, the training samples are used repeatedly for updating the model parameters in each epoch until the preset convergence criterion is reached (i.e., the absolute value of the difference in NSE between the two epochs < 0.001).

Four points need to be clarified regarding model training: (1) Regardless of local or regional simulations, we use 20 years data from October 1, 1980 to September 30, 2000 to train the model, and evaluate the model performance from October 1, 2000 to September 30, 2010. (2) To avoid the impact of local PRNN-LSTM and regional PRNN-EA-LSTM when starting in the initial state of $t = t_0$ on the performance of the entire evaluation period, we use the previous year of the evaluation period as the spin-up period of the model. Specifically, to evaluate the model performance from January 10, 2000 to September 30, 2010, the spin-up period is set from January 10, 1999 to September 30, 2000; (3) The training loss function was the average NSE over all training catchments; this is a squared-error loss function that, unlike a more traditional MSE loss function, does not overweight catchments with larger mean streamflow values (i.e., does not overweight large, humid catchments) (Kratzert et al., 2019a,b); (4) For the regional simulation, out-of-sample testing was done by k-fold validation, which splits the 531 catchments randomly into $k = 12$ groups of approximately equal size, uses all catchments from $k-1$ groups to train the model, and then validated the model on the single group of holdout catchments. This procedure is repeated $k = 12$ times so that out-of-sample predictions are available from every catchment.

3.2.2. Implementation for runoff simulation

In this study, a total of 10 models were trained and evaluated for local and regional simulations:

1. Local PRNN-LSTM: with 5 meteorological forcings and 4 physical variables of interest as inputs, trained and evaluated on each catchment individually.
2. Local PRNN: with only 5 meteorological forcings as inputs, trained and evaluated on each catchment individually.

3. Local LSTM: with only 5 meteorological forcings as inputs, trained and evaluated on each catchment individually.

4. Local EXP-HYDRO: with only 5 meteorological forcings as inputs, trained and evaluated on each catchment individually.

5. Local EXP-HYDRO-LSTM: with 5 meteorological forcings and 4 physical variables of interest as inputs, trained and evaluated on each catchment individually.

6. Regional PRNN-EA-LSTM: with 5 meteorological forcings, 4 physical variables of interest and 27 catchment attribute data as inputs, trained and evaluated on 531 catchments simultaneously with k-fold validation ($k = 12$).

7. Regional PRNN: With only 5 meteorological forcings as inputs, trained and evaluated on 531 catchments simultaneously with k-fold validation ($k = 12$).

8. Regional EA-LSTM: with 5 meteorological forcings and 27 catchment attribute data as inputs, trained and evaluated on 531 catchments simultaneously with k-fold validation ($k = 12$).

9. Regional EXP-HYDRO: With only 5 meteorological forcings as inputs, trained and evaluated on 531 catchments simultaneously with k-fold validation ($k = 12$).

10. Regional EXP-HYDRO-EA-LSTM: with 5 meteorological forcings, 4 physical variables of interest and 27 catchment attribute data as inputs, trained and evaluated on 531 catchments simultaneously with k-fold validation ($k = 12$).

Notably, model 5 and 10 solely use EXP-HYDRO (without RNN) as the input of respective data-driven cells. We need to compare its model performance with PRNN-EA-LSTM. This is important because PRNN-EA-LSTM couples the versatility of DL and conceptual hydrological models within a unified DL architecture.

4. Results and discussion

4.1. Local and regional simulations

For the local simulation, Fig. 2 presents the spatial distributions of the NSE values, for the five models during the independent evaluation period (1 October 2000 to 30 September 2010). Overall, local PRNN-LSTM outperforms the other four models. Visually, local PRNN, local EXP-HYDRO and local LSTM exhibit less satisfactory NSE values for catchments in the central, southern and southwestern United States. Table B2 shows the proportion of catchments with $NSE < 0.55$ in the three models accounts for 70.2 %, 80.4 % and 45.9 %. These catchments have experienced prolonged periods of aridity and lack of rainfall (according to the CAMELS dataset statistics, the *Low_prec_freq* (daily precipitation < 1 mm/day) > 270 days), resulting in a significant proportion of the year where runoff is equal to baseflow. In addition, flashy runoff in these catchments is primarily influenced by short-duration, high-intensity storm events, in which runoff is dominated by infiltration-excess surface flow.

Regarding the data aspect, the limited frequency of recorded data for storm events poses challenges for data-driven DL models to accurately simulate flashy runoff. In terms of physical dynamics, while local PRNN and local EXP-HYDRO, which take baseflow dynamics into account, exhibit poor performance on these catchments, it is uncertain whether DL models that do not explicitly consider baseflow dynamics can effectively capture the hydrological behavior of these catchments during rainless periods. Interestingly, local PRNN-LSTM has more favorable NSE values in these catchments, and even more than local EXP-HYDRO-LSTM. This observation highlights two key points: (1) EXP-HYDRO (with or without RNN) as the input of the data-driven cell enhances the model's understanding of the runoff process; (2) Compared to EXP-HYDRO, the inclusion of the PRNN cell in the unified DL architecture provides more valuable input to the data-driven cell. Although it is difficult to explain the reason, it can be inferred that local PRNN-LSTM potentially updates the baseflow dynamics from the baseflow sequences output by the PRNN cell. This interesting issue will be further discussed

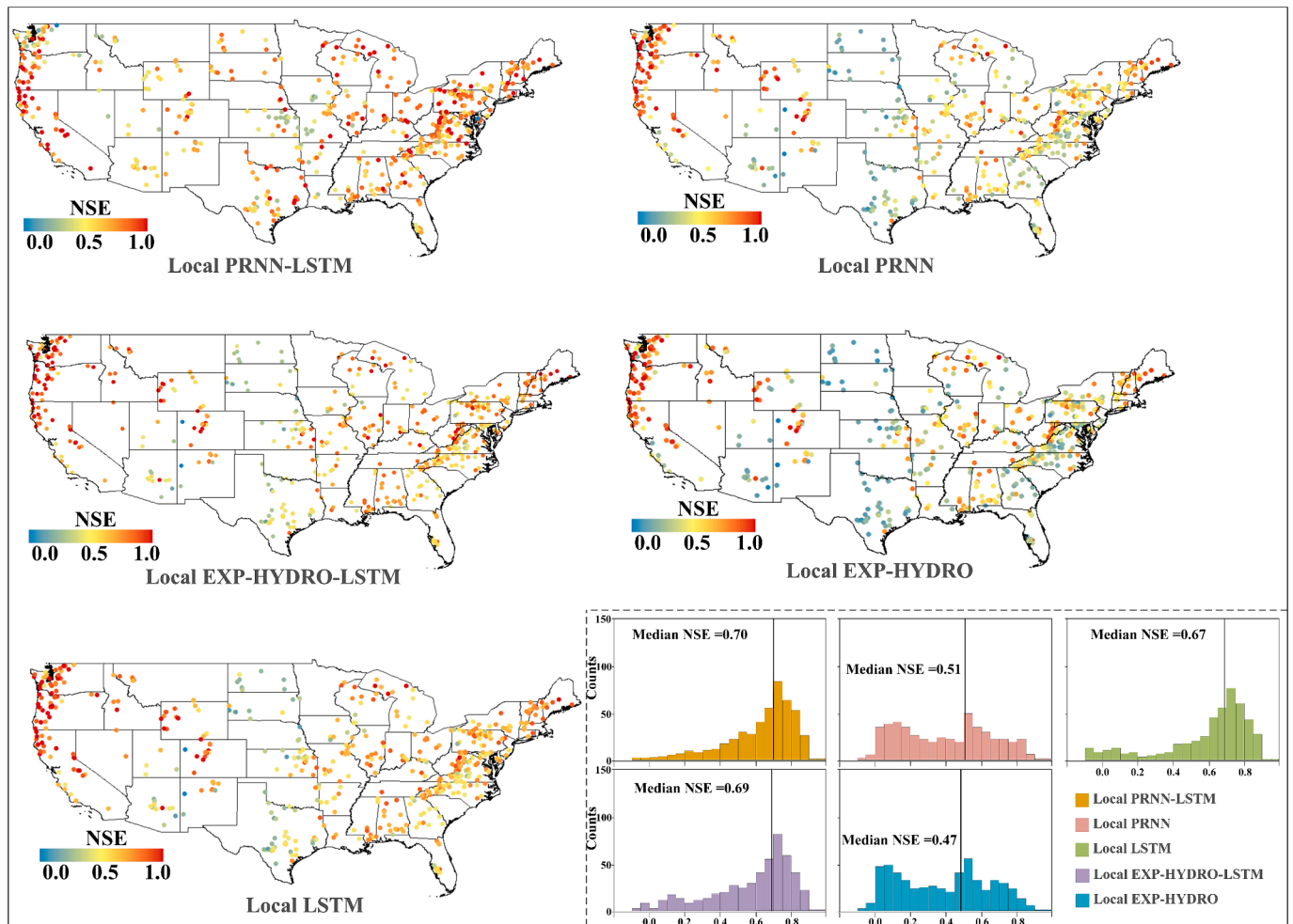


Fig. 2. The spatial distributions and histograms of the NSE for local PRNN-LSTM, local PRNN, local LSTM, local EXP-HYDRO-LSTM and local EXP-HYDRO.

in section 4.2.

For the regional simulation, Fig. 3 presents the spatial distributions and histograms of the NSE values for the five models during the evaluation period. Visually, regional PRNN-EA-LSTM, regional EA-LSTM and regional EXP-HYDRO-EA-LSTM exhibit more favorable spatial distribution of NSE values than regional PRNN and regional EXP-HYDRO, which indicates that incorporating catchment attribute data in the regional simulation can improve cross-catchment learning capabilities. Notably, regional PRNN-EA-LSTM inherits the advantages of regional EA-LSTM, which outperforms the hydrological models calibrated individually for each catchment. Although the performance of the regional EA-LSTM and regional EXP-HYDRO-EA-LSTM is slightly lower than the regional PRNN-EA-LSTM in terms of median NSE values, its simulations fluctuate significantly, and the regional EA-LSTM and regional EXP-HYDRO-EA-LSTM demonstrate notably poor predictions ($NSE < 0$) in 26 and 15 catchments (see Fig. 4). Regional PRNN-EA-LSTM achieves a higher mean NSE value and demonstrates fewer catastrophic predictions, suggesting that it possesses more robust cross-catchment learning capabilities compared to the other models. The cumulative density functions of NSE and Kling-Gupta efficiency (KGE, Gupta et al., (2009)) metrics for the regional models (Fig. 5) show the regional simulation results more explicitly.

A set of existing LSTM models and some hydrological models by several previous studies are collected over the CAMELS, which were calibrated in individual catchments or the continental United States. Table 2 shows the results of the models employed in this study alongside the results of the other models used by the researchers in the CAMELS. It

is observed that regional PRNN-EA-LSTM exhibited a higher NSE value during the evaluation period, with an average NSE of 0.67 and a median NSE of 0.72, which were 0.03 and 0.02 higher, respectively, than local PRNN-LSTM. In comparison to the LSTMs recently developed by Kratzert et al. (2019a,b), regional PRNN-EA-LSTM demonstrates superior performance. Furthermore, when compared to five process-based hydrological models, regional PRNN-EA-LSTM maintains its superiority. Table B1 provides more comparisons of other hydrological metrics.

4.2. Can PRNN-EA-LSTM be educated?

The process-driven DL model developed in this study demonstrated the best performance in the local and regional simulations, with regional PRNN-EA-LSTM achieving the highest level of performance. Regional PRNN-EA-LSTM establishes the connection between the 6 sub-process dynamics, the 5 input sequences, and the 27 catchment attribute data through the neural network. Therefore, it is difficult to quantitatively explain how the interaction among the three results in PRNN-EA-LSTM performance. Nevertheless, it is still possible to speculate on the potential factors contributing to the spatial distribution of NSE values in PRNN-EA-LSTM. This study takes regional PRNN-EA-LSTM as a case and discusses the issue.

From the perspective of DL feature importance analysis, the 531 CAMELS catchments were initially divided into 9 climate regions according to the climatologically consistent regions within CONUS as defined by the National Centers for Environmental Information (NCEI) (see Fig. 6). Subsequently, the importance of 5 input sequences, 4

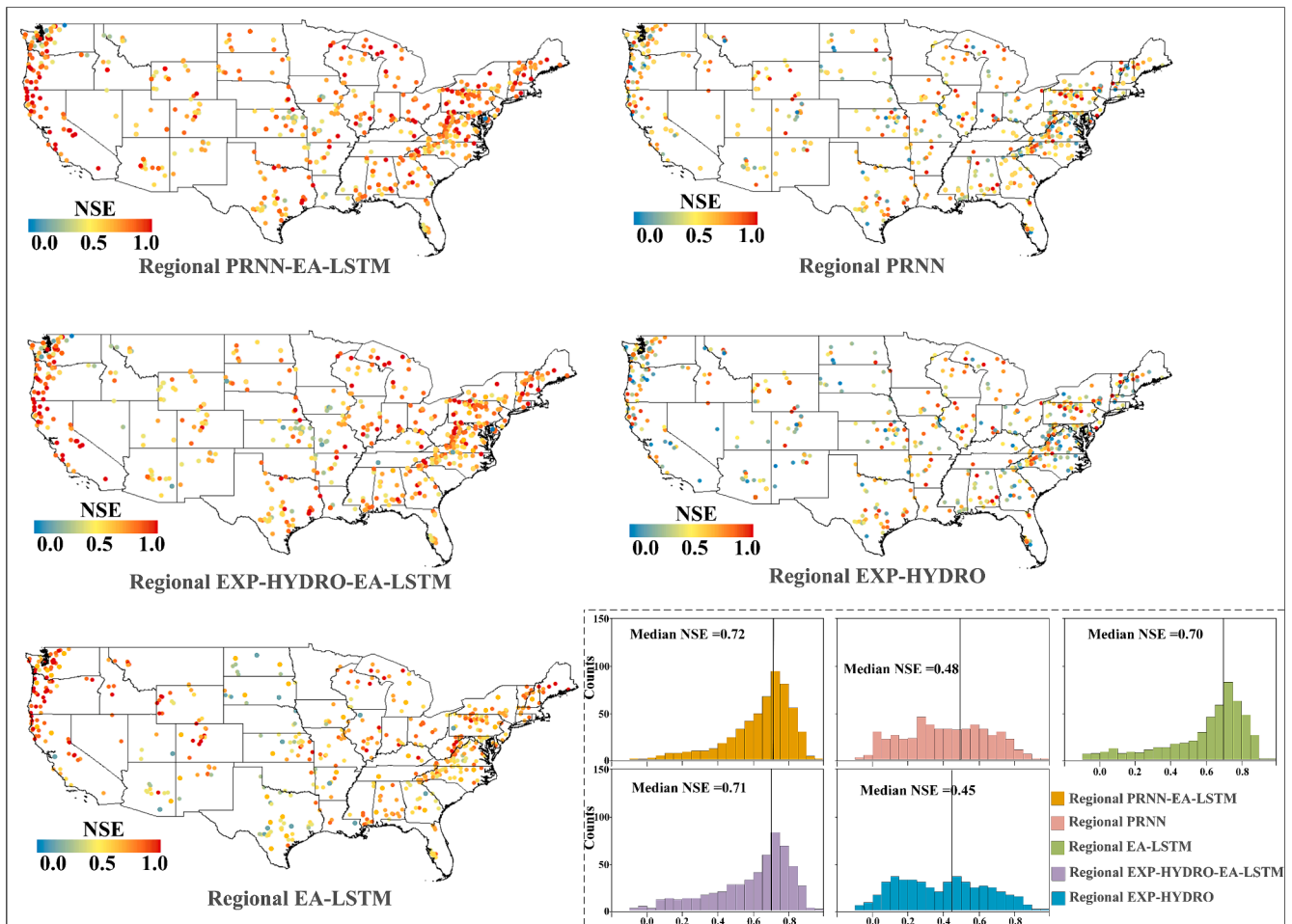


Fig. 3. The spatial distributions and histograms of the NSE for regional PRNN-EA-LSTM, regional PRNN, regional EA-LSTM, regional EXP-HYDRO-EA-LSTM and regional EXP-HYDRO.

variables of interest, and 27 catchment attribute data in relation to the performance of regional PRNN-EA-LSTM was quantified within each climate regions. As shown in Fig. 7, rainfall intensity (RI) emerges as the most important feature of each climate region, indicating its significance in estimating the runoff hydrograph of catchments. However, this importance may change when incorporating data on variables of interest and catchment attribute. The most direct evidence is the SW region, where the dominant feature is *Qbase* (i.e., baseflow variable). The climate characteristics of the SW region are primarily defined by drought and low rainfall (average *Aridity index* = 2.01, average *Low_prec_freq* = 278.12). Perennial drought and lack of rainfall mean that the runoff of the catchments in this climate region mainly depends on the baseflow supply generated by water storage. In addition, some features within climatic factors (e.g. *Aridity index*), topological factors (e.g. *Area*), or soil factors (e.g. *Soil_porosity*) also play a role in arid catchments. This again shows that catchment attribute datasets encompass valuable information regarding variations in rainfall-runoff behaviors across (eco) hydrological regimes, and that DL is effective in extracting and using this information. (The first time is that the performance of regional PRNN and regional EXP-HYDRO is inferior to that of regional PRNN-EA-LSTM, regional EXP-HYDRO-EA-LSTM and regional EA-LSTM. Interestingly, this evidence has been previously observed by Kratzert et al. (2019a,b)). Therefore, this study speculates that regional PRNN-EA-LSTM has the potential to learn new baseflow dynamics from baseflow sequences and catchment attribute data. This enables the model to better capture the hydrologic cycle of catchments under no or minimal rainfall conditions. In essence, regional PRNN-EA-LSTM demonstrates enhanced

effectiveness in simulating low flow situations. This speculation aligns with Fig. 7 of Foks et al., (2019) that high BF days (i.e., days with streamflow equal to baseflow) mainly occur in the southwestern and central lowlands.

To validate the above inference, we focus on the performance of regional PRNN-EA-LSTM in simulating low flow values, especially within catchments in the SW region. Modified NSE (mNSE, Legates and McCabe, 1999) and the Bias in the Flow Duration Curve Low-Segment Volume (FLV, Yilmaz et al., 2008) metrics were used to evaluate the low flow value simulations. As demonstrated in Table B1, regional PRNN-EA-LSTM maintains the highest mNSE value and the lowest FLV value across all catchments, which is consistent with our speculation. For the SW region, two typical arid and rainless catchments were selected for analysis. Fig. 8 displays the runoff hydrograph of the 09,505,350 catchment (*Aridity index* = 2.07, *Low_prec_freq* = 302.20), while Fig. 9 portrays the runoff hydrograph of the 09,513,780 catchment (*Aridity index* = 2.98, *Low_prec_freq* = 325.50) throughout the evaluation period. The runoff hydrograph simulated by PRNN-EA-LSTM is more physically consistent than pseudo-PRNN-EA-LSTM (without baseflow dynamics) on the number of days with no or little rain. Pseudo-PRNN-EA-LSTM oscillates significantly on days near zero observations, and instances of negative runoff values occur frequently. In contrast, the runoff values simulated by PRNN-EA-LSTM on days with no or minimal rainfall are almost consistent with the observations, and the vibration amplitude is significantly reduced. Consequently, the NSE values for the 09,505,350 and 09,513,780 catchments improved from 0.31 and 0.26 to 0.78 and 0.90, respectively. Additionally, and the FLV values decreased

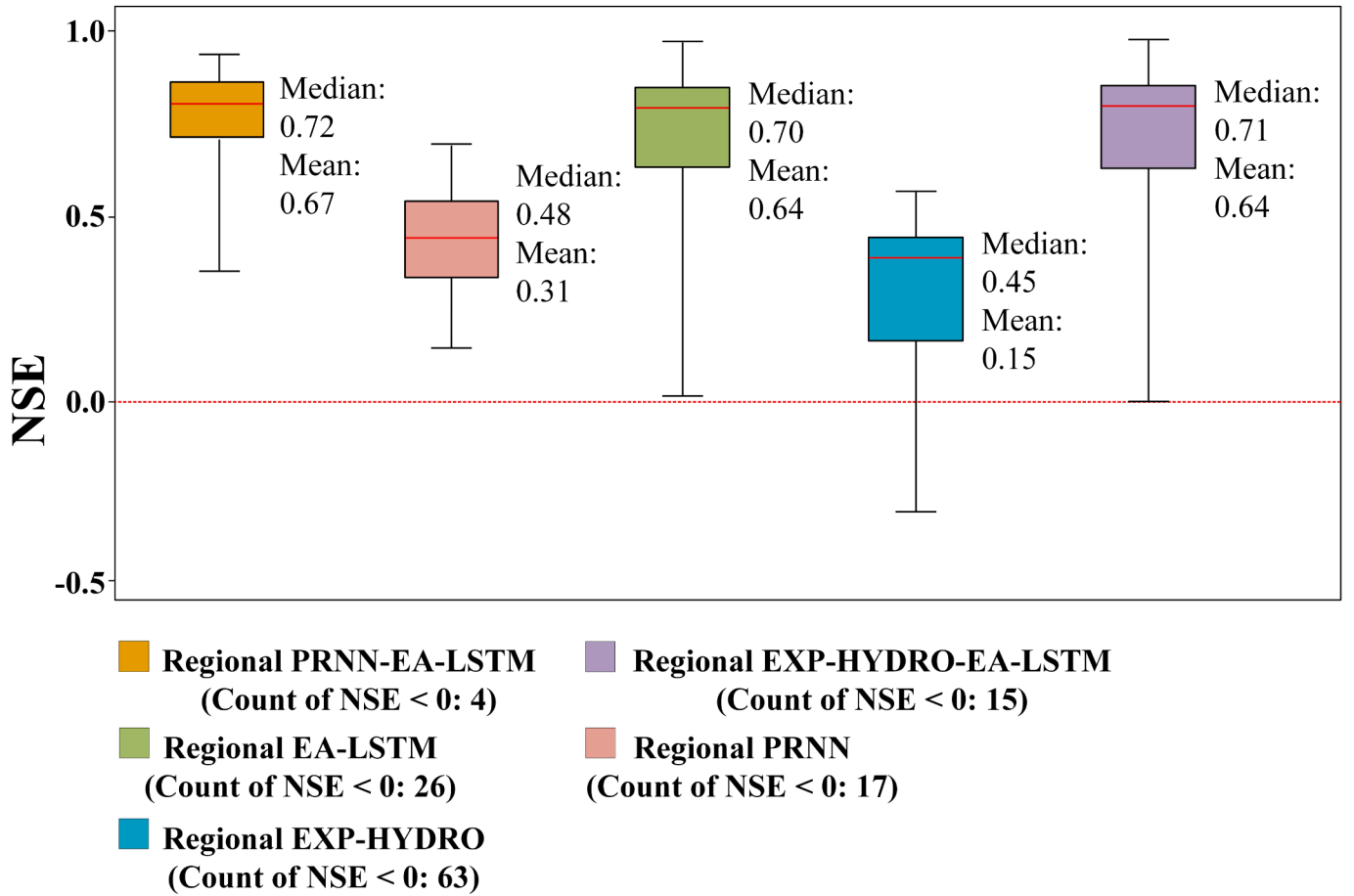


Fig. 4. NSE boxplots of the five regional models.

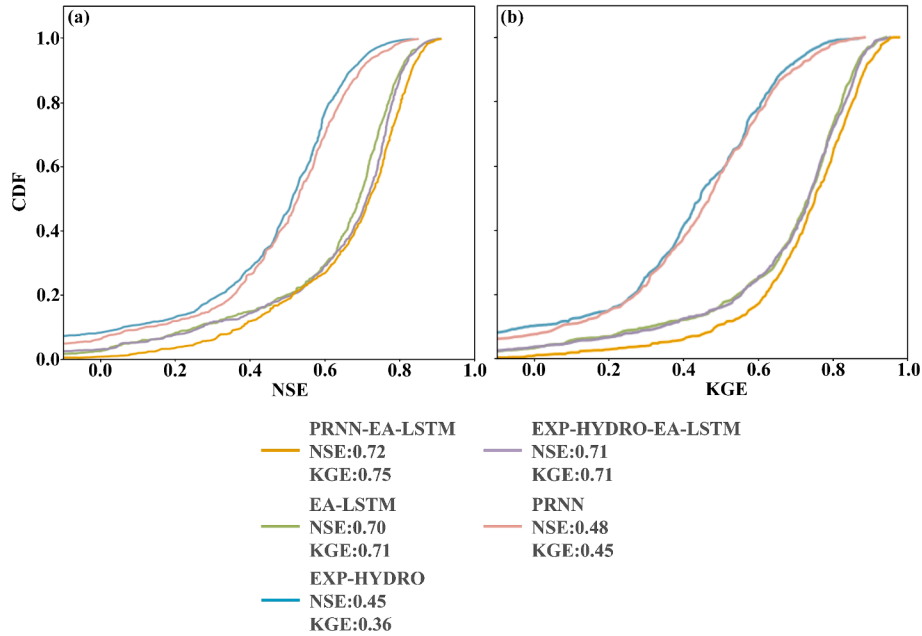


Fig. 5. Cumulative density functions of (a) NSE and (b) KGE metrics for 5 regional models during the evaluation period.

from 44.1 and 47.3 to 19.8 and 11.7, respectively.

In summary, PRNN-EA-LSTM was effectively trained using the incorporated baseflow dynamics, resulting in an improved understanding of the cross-catchment runoff process.

4.3. Synergy of physics and deep learning

Lastly, it is important to reiterate that a valuable feature of PRNN-EA-LSTM lies in the synergistic operation of its process elements and

Table 2

Comparison of NSE for the evaluation period derived from different models, either developed in this study or referred to the literature.

Model	Scale	Input	Counts	NSE		Reference
				Mean	Median	
PRNN-LSTM	Local	Without static inputs	531	0.64	0.70	This study
PRNN	Local	Without static inputs	531	0.13	0.51	This study
LSTM	Local	Without static inputs	531	0.61	0.67	This study
EXP-HYDRO	Local	Without static inputs	531	0.08	0.47	This study
EXP-HYDRO-LSTM	Local	Without static inputs	531	0.61	0.69	This study
PRNN-EA-LSTM	Regional	With static inputs	531	0.67	0.72	This study
PRNN	Regional	Without static inputs	531	0.31	0.48	This study
EA-LSTM	Regional	With static inputs	531	0.64	0.70	This study
EXP-HYDRO	Regional	Without static inputs	531	0.15	0.45	This study
EXP-HYDRO-EA-LSTM	Regional	Without static inputs	531	0.64	0.71	This study
LSTM	Regional	With static inputs	531	0.54	0.69	Kratzert et al. (2019a,b)
LSTM	Regional	Without static inputs	531	0.45	0.63	Kratzert et al. (2019a,b)
EXP-HYDRO RNN	Regional	With static inputs	569	0.58	0.66	Jiang et al. (2020)
EXP-HYDRO RNN	Local	Without static inputs	569	0.62	0.71	Jiang et al. (2020)
XAJ ¹	Local	–	531	0.47	0.52	Xie et al. (2021)
VIC ²	Local	–	531	0.51	0.55	Newman et al. (2017)
VIC	Regional	–	531	0.17	0.31	Mizukami et al. (2019)
SAC-SMA ³	Local	–	241	0.56	0.60	Newman et al. (2017)
HBV ⁴	Local	–	447	0.43	0.54	Seibert et al. (2018)
FUSE ⁵	Local	–	447	0.58	0.62	Kratzert et al. (2019a,b)

1. Xinjiang model; 2. Variable infiltration capacity model; 3. Sacramento soil moisture accounting model; 4. Hydrologiska Byråns Vattenbalansavdelning model; 5. Framework for understanding structural error model.

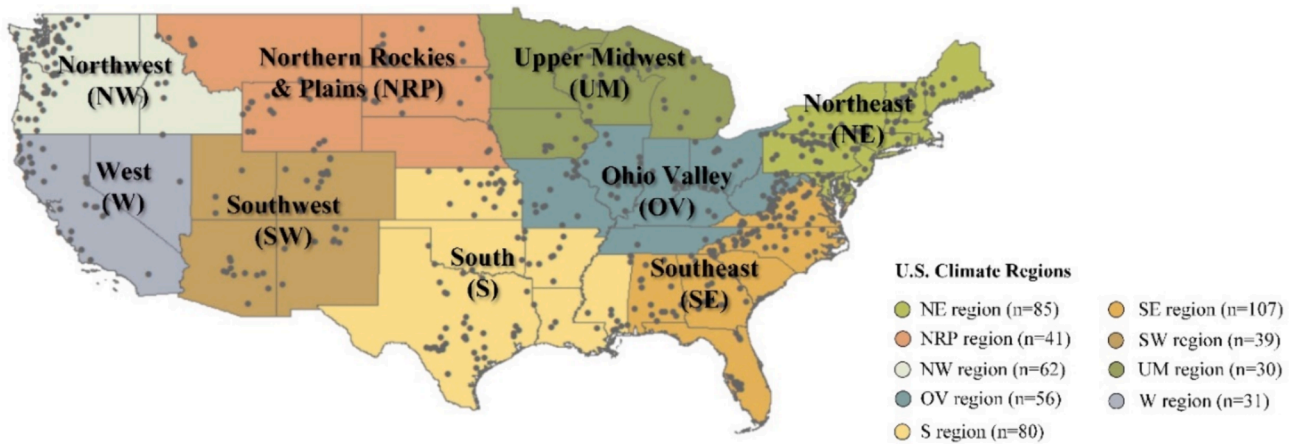


Fig. 6. Map displaying gauge locations according to NOAA's climatologically similar regions throughout the continental United States. US climate regions identified by the National Climate Data Center ().

adapted from Karl and Koss, 1984, <https://www.ncdc.noaa.gov/monitoring-references/maps/us-climate-regions.php>

DL elements within a unified architecture. Thus, it is essential to examine whether this feature of PRNN-EA-LSTM is evident in the results. To this end, the relative changes in the total runoff output by a PRNN cell and PRNN-EA-LSTM during the evaluation period are compared (see Fig. 10(a)).

In most catchments, the relative changes in total runoff simulated by the two models were not substantially different (<20 %). However, notable changes were concentrated in the central Great Plains and Southwest, as well as in certain catchments near the Appalachian Mountains and the Pacific Northwest. It can be seen from the relationships between the relative change of total runoff and the Aridity index and Low_prec_freq in the catchment (Fig. 10(a)) that most of these catchments fall within the subset characterized by $Aridity\ index \geq 1.1$ and $Low_prec_freq \geq 270$. This observation indicates two possibilities: (1) the PRNN cell encounters unknown catchment behaviors that cannot be resolved analytically, or (2) the data-driven cell causes PRNN-EA-LSTM to overfit. Nevertheless, there is an expectation that PRNN-EA-LSTM is likely to perform satisfactorily in such scenarios. To verify this expectation, the NSE of PRNN-EA-LSTM is compared to that of the PRNN cell

specifically within these catchments.

Fig. 10(b) illustrates the NSE difference between PRNN-EA-LSTM and PRNN cell for the aforementioned catchment collection (a total of 102 catchments). The implication of this result is that when the PRNN cell exhibits poor performance in some catchments (in the central Great Plains region and the Southwest of the United States), the data-driven cell takes precedence in the model simulation and discovers new catchment behaviors that the PRNN cell can not simulate. This also implies that the physical understanding of the PRNN cell for the catchment behaviors (i.e., EXP-HYDRO) has yet to be perfected. On the other hand, the PRNN cell performs well in non-arid catchments, where it dominates the simulations, while the data-driven cell fine-tunes the hydrograph to ensure a balanced water budget. Although the data-driven cell may introduce slight perturbations to the final outputs, PRNN-EA-LSTM does not fluctuate as significantly as pure data-driven DL models in mean and median NSE from the known catchments to the unknown catchments. Future work will rigorously evaluate this hypothesis.

The above results conclusively demonstrate that PRNN-EA-LSTM

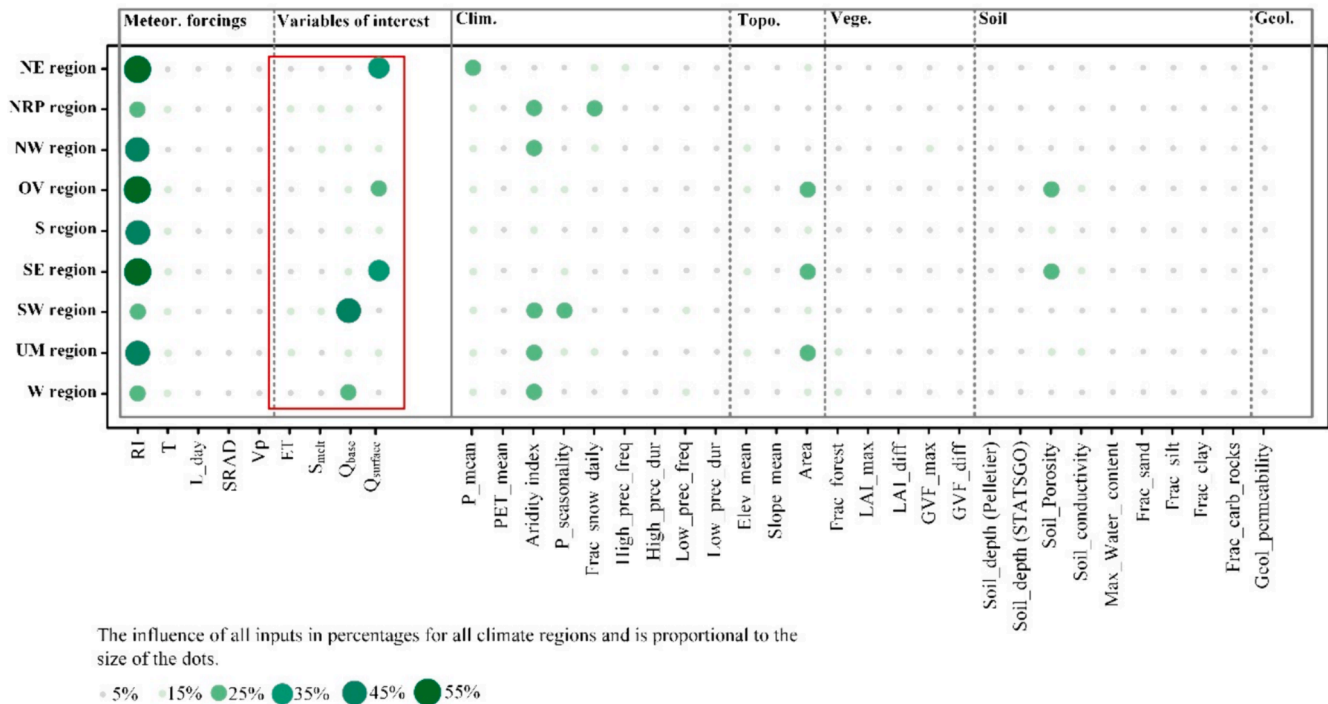


Fig. 7. Comparison of the influence of all features (x axis) on runoff of catchment clusters in each climate region (y axis) in regional PRNN-EA-LSTM by using the Permutation method.

successfully attains its designated objective. The synergy between physics and DL within a unified architecture has yielded an effect where the combination is greater than the sum of its parts.

5. Conclusions

In this study, a process-driven deep learning (DL) model (PRNN-EA-LSTM) was proposed to improve the process understanding capability of pure DL models. The findings demonstrate that incorporating process-based models as inputs within the DL architecture significantly improves the model's comprehension of underlying processes and enhances the robustness of simulations. In the PRNN-EA-LSTM model architecture, process-based component contribute to the process understanding of hydrological systems, and data-driven component fill the knowledge gaps of process-based component. Conclusions derived are as follows:

- (1) A large sample dataset CAMELS was used as the study case, PRNN-EA-LSTM demonstrated performance close to state-of-the-art models in local and regional simulations. Specifically, the median NSE of PRNN-EA-LSTM for local and regional simulations was 0.70 and 0.72, respectively, while the mean NSE was 0.64 and 0.67, respectively. Regional PRNN-EA-LSTM exhibited fewer catastrophic predictions compared to process-based models and pure DL models, showcasing its robust cross-catchment learning capabilities.
- (2) The baseflow sequences output by the PRNN cell effectively educate PRNN-EA-LSTM, enabling it to improve its ability to simulate low flow values. PRNN-EA-LSTM successfully preserves the normal hydrological cycle of arid catchments during extended periods of minimal or no rainfall, effectively mitigated the occurrence of negative flows.
- (3) The synergy between the process-based component (i.e., PRNN cell) and the data-driven component (i.e., EA-LSTM cell) was the key to the success of PRNN-EA-LSTM. Specifically, PRNN-EA-LSTM effectively enhanced the NSE of 102 catchments

exhibiting significant drought characteristics (*Aridity index* ≥ 1.1 and *Low_prec_freq* ≥ 270).

Regarding future applications in rainfall-runoff simulation, a key next step would be to encode more runoff mechanisms and process dynamics (e.g., flow in soil) in PRNN-EA-LSTM. Including process dynamics within the DL architecture brings another potential advantage: the ability to use multiple observations to constrain the model's learning of the target variable. For example, with the availability of dense ground monitoring sites that accurately record observations of various hydrological variables, integrating additional observations to constrain different sub-process dynamics of the model can effectively reduce uncertainty and enhance the model's robustness (Dembélé et al., 2020; Efstratiadis and Koutsoyiannis, 2010).

CRedit authorship contribution statement

Heng Li: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Chunxiao Zhang:** Writing – review & editing, Supervision, Resources, Methodology, Funding acquisition, Conceptualization. **Wenhao Chu:** Writing – original draft, Methodology, Data curation. **Dingtao Shen:** Writing – review & editing, Resources, Funding acquisition, Conceptualization. **Rongrong Li:** Writing – review & editing, Supervision, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

CAMELS data can be downloaded at <https://doi.org/10.5065/>

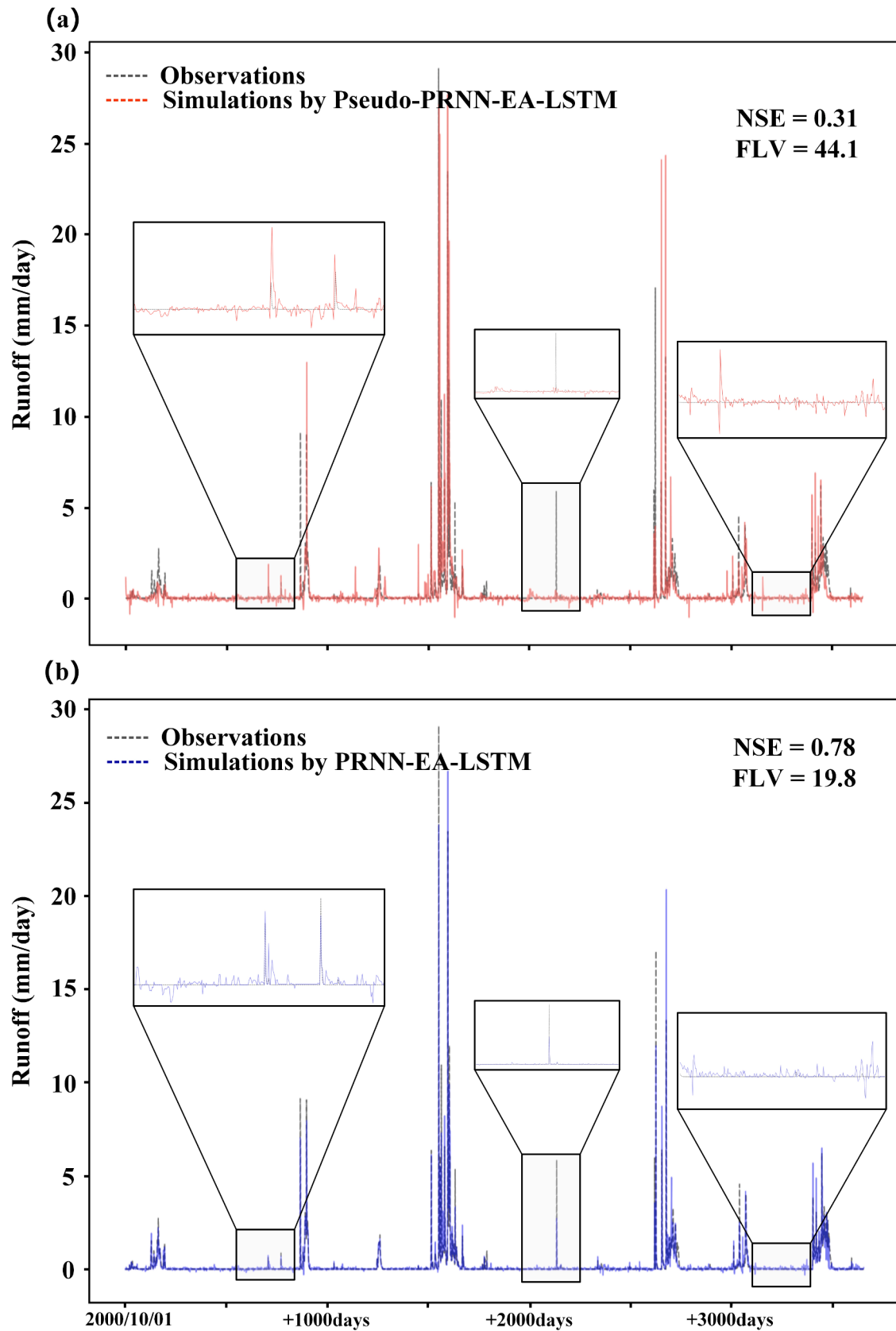


Fig. 8. Comparison of the runoff hydrographs simulated by PRNN-EA-LSTM and pseudo-PRNN-EA-LSTM during the evaluation periods. (a) Runoff hydrograph of Pseudo-PRNN-EA-LSTM (without baseflow dynamics) in 09,505,350 catchment during the evaluation period. (b) Runoff hydrograph of PRNN-EA-LSTM (with baseflow dynamics) in 09,505,350 catchment during the evaluation period.

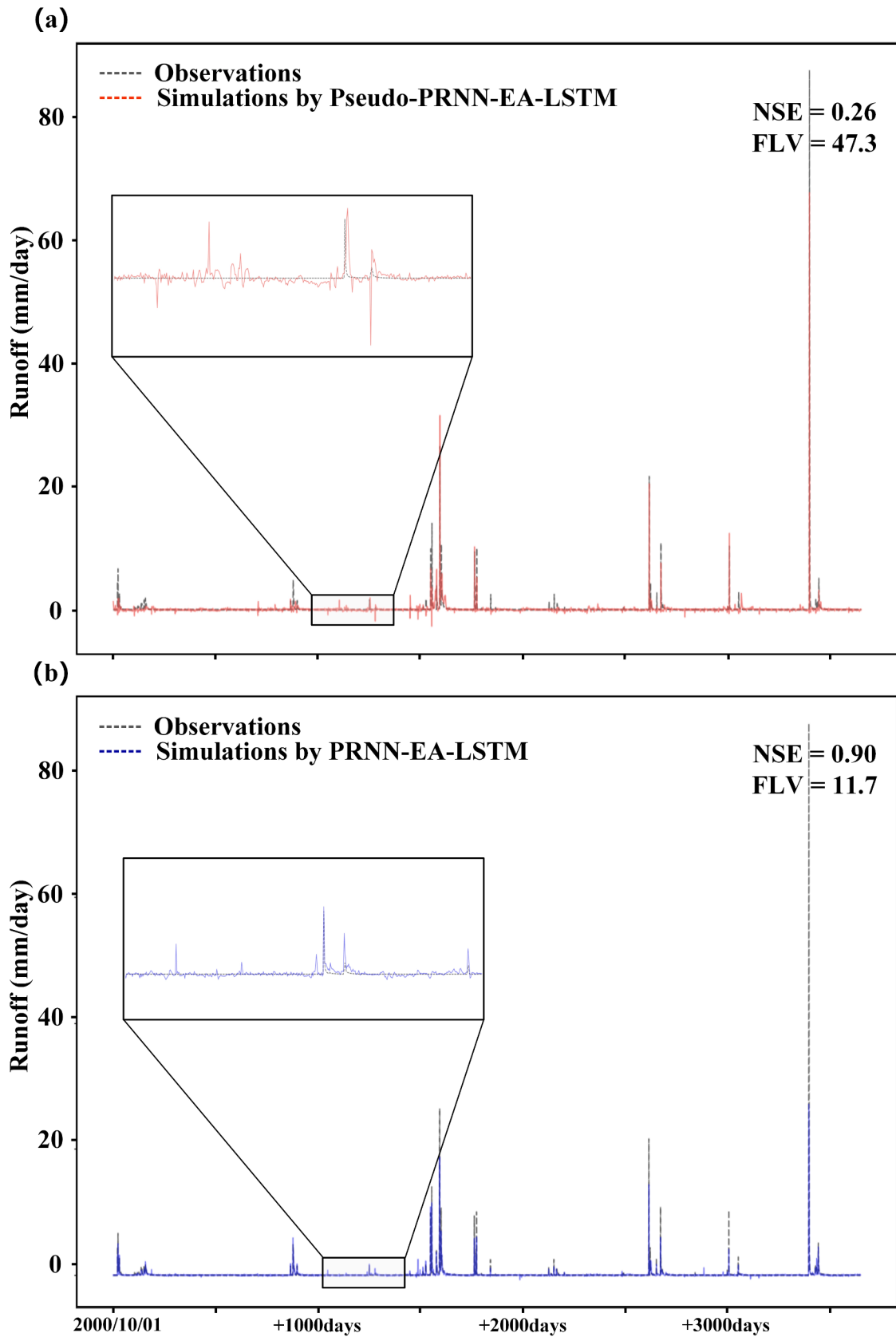


Fig. 9. Comparison of the runoff hydrographs simulated by PRNN-EA-LSTM and pseudo-PRNN-EA-LSTM during the evaluation periods. (a) Runoff hydrograph of Pseudo-PRNN-EA-LSTM (without baseflow dynamics) in 09,513,780 catchment during the evaluation period. (b) Runoff hydrograph of PRNN-EA-LSTM (with baseflow dynamics) in 09,513,780 catchment during the evaluation period.

(a) Relationship between the relative change (absolute value) of total runoff during the evaluation period and *Aridity index* and *Low_prec_freq* all over the catchments

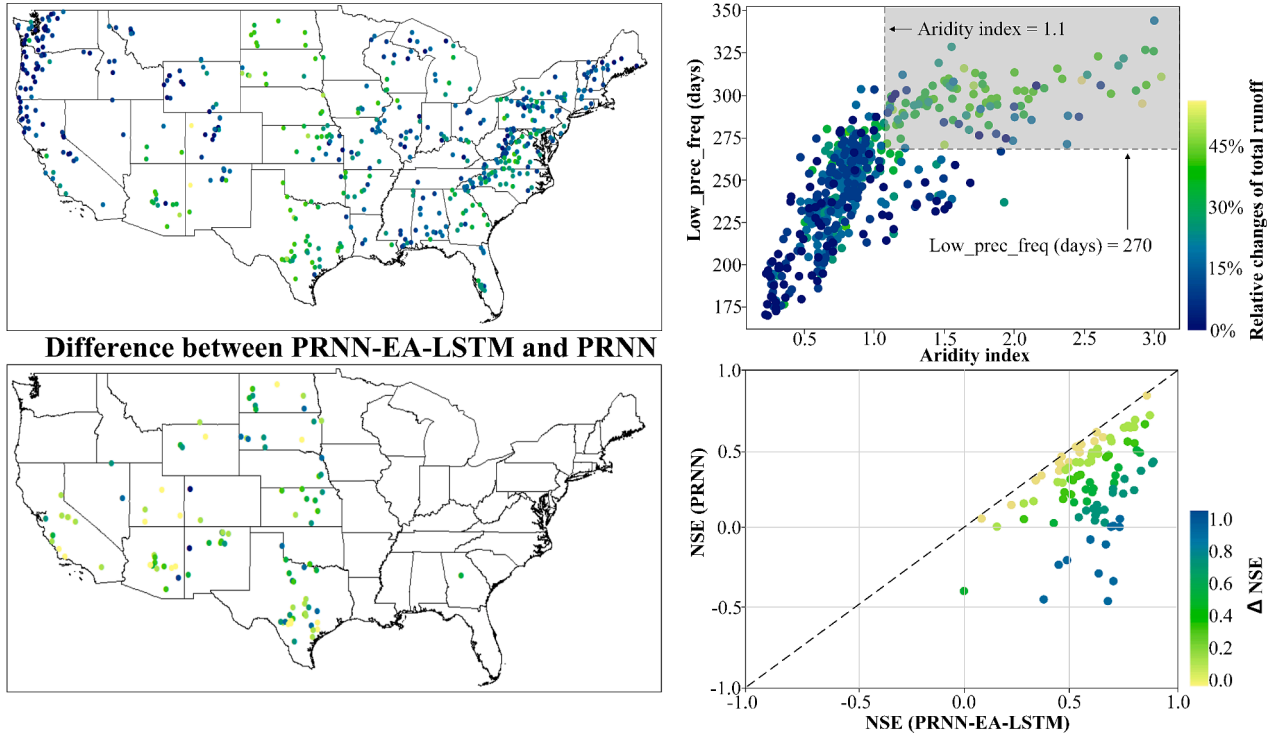


Fig. 10. The synergy of physics and DL in PRNN-EA-LSTM. (a) Relative changes (in absolute values) in the total runoff during the evaluation period, comparing the final output runoff Q of PRNN-EA-LSTM against the respective intermediate estimated Q (i.e., $Q_{base} + Q_{surface}$) of the PRNN cell. (b) Comparison between PRNN-EA-LSTM and PRNN under the condition of *Aridity index* ≥ 1.1 and *Low_prec_freq* ≥ 270 .

D6MW2F4D (Addor et al., 2017). The code used for PRNN-EA-LSTM is publicly available (<https://github.com/lhmygis>).

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Appendix A

A1 EXP-HYDRO model

In EXP-HYDRO, only the precipitation that is considered as snowfall accumulates in the snow accumulation bucket, whereas the rainfall accumulates directly in the catchment bucket. The observed daily precipitation P (mm/day) is classified as P_s or P_r depending on the following conditions:

If observed daily air temperature T ($^{\circ}\text{C}$) $\leq T_{min}$,

$$P_s = P$$

$$P_r = 0$$

Else,

$$P_s = 0$$

$$P_r = P$$

where T_{min} is an air temperature threshold, below which precipitation is in the form of snowfall.

The snowmelt M is modelled using the thermal degree-day concept related to T and the storage in the snow accumulation bucket S_0 , as follows:

If $S_0 > 0$ and $T > T_{max}$,

$$M = \min\{S_0, D_f \cdot (T - T_{max})\}$$

Else,

$$M = 0$$

where D_f is the thermal degree-day factor (mm/day/°C). T_{max} is the air temperature threshold above which accumulated snow begins to melt. The snowmelt M from the snow accumulation bucket is input to the catchment bucket.

The evapotranspiration ET is calculated as a fraction of the potential evapotranspiration PET (mm/day), which depends on the current storage in the catchment bucket $S1$ relative to the storage capacity of the catchment bucket S_{max} , as:

$$\text{If } S1 < 0,$$

$$ET = 0$$

$$\text{Else if } 0 \leq S1 \leq S_{max},$$

$$ET = PET \cdot (S1/S_{max})$$

$$\text{Else if } S_{max} < S1,$$

$$ET = PET$$

The PET is estimated by Hamon's formulation:

$$PET = 29.8 \cdot L_{day} \cdot \frac{e_{sat}(T)}{T + 273.2}$$

where L_{day} is the day length (hour), which depends on the Julian date of the year and the latitude of catchment location. The $e_{sat}(T)$ is the saturation vapor pressure (kPa) approximated as:

$$e_{sat}(T) = 0.611 \cdot \exp\left(\frac{17.3 \cdot T}{T + 237.3}\right)$$

The runoff at the catchment outlet Q_{sum} , referred to the TOPgraphy based hydrological MODEL (i.e., TOPMODEL), is modeled as the sum of the baseflow (Q_{base}) generated depending on the available storage in the catchment bucket and the capacity-excess runoff ($Q_{surface}$) generated when catchment bucket is saturated, which are expressed as:

$$\text{If } S1 < 0,$$

$$Q_{base} = 0$$

$$Q_{surface} = 0$$

$$\text{Else if } 0 \leq S1 \leq S_{max},$$

$$Q_{base} = (1-f) Q_{max} \cdot e^{-f(S_{max} - S1)}$$

$$Q_{surface} = 0$$

$$\text{Else if } S_{max} < S1,$$

$$Q_{base} = Q_{max}$$

$$Q_{surface} = S1 - S_{max}$$

$$Q_{sum} = Q_{base} + Q_{surface}$$

where f is the parameter controlling the storage-dependent decline in runoff (mm⁻¹) and Q_{max} is the maximum subsurface runoff produced by the catchment bucket (mm/day).

Table A1

EXP-HYDRO parameter definitions, meaning, units and value range.

Parameters	Description	Units	Lower limit	Upper limit
f	the rate of decline in runoff from catchment bucket	mm ⁻¹	0.0	0.1
S_{max}	Maximum storage of the catchment bucket	mm	100.0	1500.0
Q_{max}	Maximum subsurface runoff at full bucket	mm/day	10.0	50.0
D_f	Thermal degree-day factor	mm/day/°C	0.0	5.0
T_{max}	Temperature above which snow starts melting	°C	0.0	3.0
T_{min}	Temperature below which precipitation is snow	°C	-3.0	0.0

Table A2

The computation logic (pseudocode) of the PRNN cell.

Algorithm 1: PRNN recurrent solver function

Inputs: Initial network parameters (e.g., $T_{\min}$, D_f , $T_{\max}$), sequences data (e.g., P , T), as well as $S0$ and $S1$ state variables

1: procedure function PRNN (states, sequences, parameters)

2: self.weight = parameters

3: self.S0, self.S1 = states[0], states[1]

4: function RNN_step_do (self.weight, self.S0, self.S1, sequences)

5: calculate P_s , P_r , M , ET , Q_s and Q_b refer to Appendix A1.

6: calculate next step $S0$ and $S1$ using hydrologic dynamic equations.

7: return next step $S0$ and $S1$

8: Output: The sequence of the variables of interest Q_s , Q_b , ET and M

Table A3

The computation logic (pseudocode) of the EA-LSTM cell.

Algorithm 2: EA-LSTM recurrent solver function

Inputs: Initial dynamic sequences data (e.g., P , T , Q_s and Q_b), static attribute data (e.g. topography, vegetation, soil attributes)

1: procedure function EA-LSTM (dynamic_sequences, static_sequences)

2: $i = \text{sigmoid}(\text{dot_add}(\text{calculate static_sequences}))$

3: function step_do (dynamic_sequences)

4: step initial h_0 , c_0

5: f , o , $g = \text{dot_add}(\text{calculate dynamic_sequences})$

6: next_c_state = $\text{sigmoid}(f) * c_0 + i * \tanh(g)$

7: next_h_state = $\text{sigmoid}(o) * \tanh(\text{next_c_state})$

8: $h_n.append(\text{next_h_state})$

9: $c_n.append(\text{next_c_state})$

10: Output: return the sequence of h_n , c_n

Appendix B. The evaluation and comparison results of different models developed in this study**Table B1**

The three evaluation metrics results of all the models. Note: NSE, KGE, mNSE and FLV metrics are median values.

Model	NSE	KGE	mNSE	FLV
Local PRNN-LSTM	0.70	0.72	0.54	27.3
Local PRNN	0.51	0.44	0.38	34.5
Local LSTM	0.67	0.64	0.50	29.6
Local EXP-HYDRO	0.47	0.35	0.33	36.7
Local EXP-HYDRO-LSTM	0.69	0.66	0.51	27.7
Regional PRNN-EA-LSTM	0.72	0.75	0.57	25.7
Regional PRNN	0.48	0.45	0.35	35.7
Regional EA-LSTM	0.70	0.71	0.52	28.4
Regional EXP-HYDRO	0.45	0.36	0.29	37.2
Regional EXP-HYDRO-EA-LSTM	0.71	0.71	0.53	27.1

Table B2

Catchment proportions according to their respective performances resulting from all the models developed in this study.

Model	$NSE \geq 0.55$	$0 < NSE < 0.55$	$NSE < 0$
Local PRNN-LSTM	76.8 %	21.9 %	1.3 %
Local PRNN	29.8 %	67.4 %	2.8 %
Local LSTM	54.1 %	37.6 %	8.3 %
Local EXP-HYDRO	19.6 %	71.0 %	9.4 %
Local EXP-HYDRO-LSTM	58.4 %	34.8 %	6.8 %
Regional PRNN-EA-LSTM	79.7 %	19.55 %	0.75 %
Regional PRNN	27.5 %	69.3 %	3.2 %
Regional EA-LSTM	64.3 %	30.8 %	4.9 %
Regional EXP-HYDRO	21.5 %	66.7 %	11.8 %
Regional EXP-HYDRO-EA-LSTM	67.5 %	29.7 %	2.8 %

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